

Is the 2021 E-Cigarette Entry Wave Real? A Coverage-vs-Entry Check

Verifying the 7.2a hypothesis before specifying the competitive fringe's arrival process. Companion to o8_27_2026_ecig_firm_dynamics_diagnostics and Appendix B of the brand-dynamics note.

September 1, 2026

1. What this note does, and the bottom line

Diagnostic 7.2a found that of the 63 brand-codes with a first tracked-sales month after January 2013, **27 arrive in calendar 2021** — against a background rate of 0–7 per year in every other year — and hypothesised that this is mostly the RMS panel starting to see modern disposable brands that were already selling through channels it covers poorly, not genuine de novo entry. This note checks that against the data on five margins. The fringe's entry process in the model depends on the answer: a coverage/taxonomy artifact should be corrected out of the entry margin, whereas genuine burst entry needs an arrival process that can produce a burst.

Bottom line. The 2021 wave is **almost entirely a data-visibility event, not economic entry** — and, importantly, *not* the "RMS started covering vape shops" version of that story. The RMS store panel did **not** expand in 2021; it had already contracted by ~half in early 2020 and kept shrinking. What changes at the January 2021 seam is that the new Nielsen data delivery **resolves ~110 more disposable UPCs and ~15 more brand-codes on the same (slightly smaller) set of convenience/gas/drug stores**, with total tracked units essentially flat. The products were physically on those shelves in 2020 — the panel just wasn't attributing them to brands (the residual "UNKNOWN" bucket, ~0.3–0.6M mL/month pre-2021, drops to zero in January 2021).

For the fringe process, split the 27 by *how* they appear, since both are coverage but they need different handling:

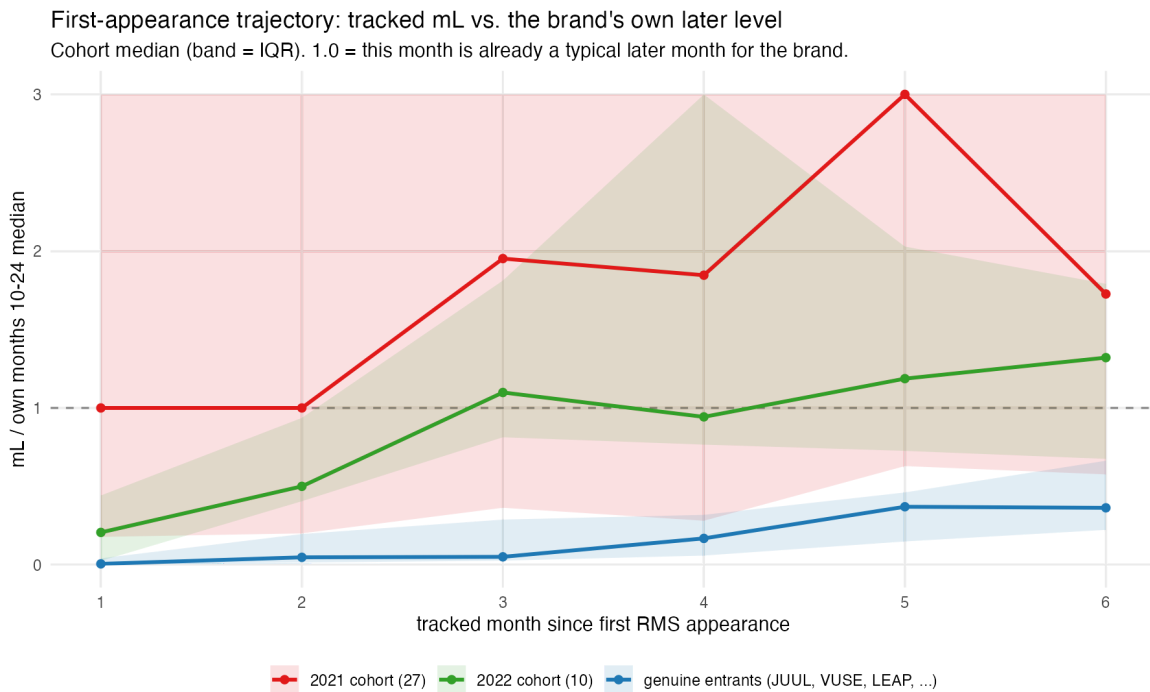
- **~18–20 of 27 — step appearance / already at scale.** Brands that show up in January 2021 already spanning many states (MNGO in 34 states, Bidi in 21) at or above their eventual steady-state volume, most with documented pre-2021 market presence, several being the *same physical product* under multiple new codes (the Puff Bar family; Puur Vapor/Savage). These are not arrivals at all — treat as a one-time panel-onboarding event at January 2021.
- **~7–9 of 27 — coverage ramp.** Brands that also existed before 2021 (Hype, Hyde, HQD, Air Bar, ...) but phase into the panel gradually over 2021 from a single state. The ramp is *coverage*

diffusion, not *product* diffusion — still not economic entry, but it looks like entry in the series.

- **~0–2 of 27 — verifiably genuine de novo 2021 entry.** On the evidence here, essentially none of the 27 can be confirmed as a product that did not exist before 2021. The cleaner genuine-entry ramps are in the 2022 cohort (Esco Bars, Fume, Avata — single-state launches that build over months).

The timing seals it: **zero** new brand-codes appear in the ten months after the February 2020 flavored-cartridge enforcement or the four months after the September 2020 PMTA deadline, then **15 appear in January 2021**, exactly at the seam where the RMS e-cig series switches Nielsen data deliveries (nielsen_extracts/RMS → 2021-Onward_Scanner_Data/...). Regulation-driven burst entry would have shown up *in 2020*.

2. First-appearance trajectories: abrupt onset vs. diffusion ramp



Cohort-median tracked mL in RMS-months 1–6, as a fraction of the brand's own months-10–24 median. 1.0 (dashed) = the first month is already a typical later month. Band = IQR.

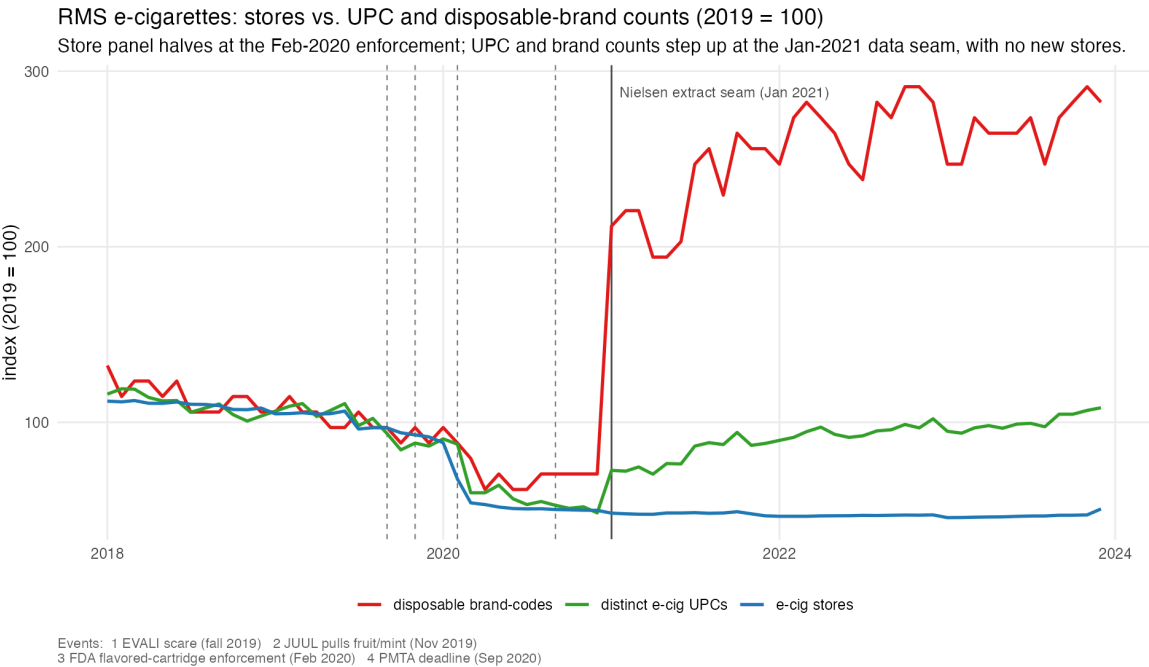
For each brand, compare its tracked mL in months 1–3 of its RMS life against its own later steady-state level (median mL over tracked-months 10–24). A genuine new product ramps: month 1 is a small fraction of where it settles. An already-established product entering measured channels appears at once near — or above — its steady state.

cohort	n	median (month-1 mL ÷ own later level)	median month-1 national mL share
clean genuine entrants (JUUL '15, VUSE '13, Logic '13, MarkTen '13, LEAP '19)	5	0.01	~0.00%
2013 post- January cohort	19	0.38	0.06%
2019 cohort	3	0.07	0.00%
2021 cohort	27	1.00	0.00%
2022 cohort	10	0.21	0.00%

The 2021 cohort's median brand records as much volume in its *first* tracked month as in a typical later month — two orders of magnitude more front-loaded than a genuine entrant, and well above even the (already coverage-affected) 2013 cohort. Several 2021 entrants come in *above* their eventual level: Bidi (month 1 = 9× later median), EPIC (21×), Dinner Lady (6×), Pop (4×), MNGO (3.6×), Switch Mods (4.7×) — an unambiguous signature of a product that was already at scale and got switched on, then partially settled as coverage stabilised. By contrast the 2022 cohort (Esco Bars, Fume, Avata, Stagbar, Savage) ramps far more gradually (median 0.21), so 2022 entry already looks closer to real diffusion.

Within 2021 the picture splits by *shape*, not by genuine-vs-fake: abrupt onset at or above steady state (Bidi, MNGO, Pop, Puur Vapor, EPIC, Dinner Lady, Switch Mods, Helo, and most of the long tail) versus multi-month ramp from near zero (Hyppe 0 → 4.3% peak, Ignite, HQD, Zeo, Pacha, Hyde). Both groups are made up of products with pre-2021 market presence (§4); the ramp is the panel's coverage of that brand improving gradually rather than the product diffusing.

3. Outlet coverage: did the store panel expand, or just the brand count?



RMS e-cig stores, distinct e-cig UPCs, and disposable brand-codes, indexed to 2019 = 100. Numbered dashed lines are the EVALI / regulatory events; the solid line is the January-2021 Nielsen data-delivery seam.

No. The count of RMS stores selling e-cigarettes **fell** across the whole relevant window, and the January 2021 brand-count jump happens on a *shrinking* store base.

month	e-cig stores	disposable-carrying stores	distinct e-cig UPCs	e-cig units
2019 (mean)	25,800	22,400	463	5.6M
2020-01	22,802	17,167	419	4.6M
2020-02	17,471	13,501	405	5.1M
2020-03	13,963	11,958	277	4.1M
2020-12	12,861	10,568	224	4.6M

month	e-cig stores	disposable-carrying stores	distinct e-cig UPCs	e-cig units
2021-01	12,450	11,328	336	5.7M
2021-06	12,475	11,278	353	4.5M
2022 (mean)	12,087	11,194	439	4.2M

Two things stand out. First, the store panel **collapses in February–March 2020** (22,800 → 14,000 e-cig stores; distinct UPCs 405 → 277), exactly at the flavored- cartridge enforcement — thousands of convenience outlets dropped the flavored SKUs (this is the enforcement's footprint in the data, worth noting separately). It keeps drifting down through 2021–2023. Second, across the December 2020 → January 2021 seam the store count goes **down** (12,861 → 12,450; 12,075 in both months, 375 new, 786 dropped — ordinary panel churn), while the distinct e-cig UPC count jumps **+50% (224 → 336)** and disposable brand-codes go 9 → 25, with tracked units flat. More products, more brands, fewer stores, same volume: that is a resolution/attribution change in the data feed, not new outlet coverage.

Where the volume comes from is consistent with this: the residual **UNKNOWN** brand bucket carries ~0.3–0.6M mL/month through 2020 and then reads **exactly 0.00 from January 2021 on**, as the new feed attributes everything to a named brand; measured disposable volume steps from ~0.3M to ~1.5M mL/month at the same instant.

The entrants' first-month geographic footprint is the useful discriminator for §7. Brands appearing all-at-once across many states in month 1 — MNGO (34 states), Bidi (21), Dinner Lady (8) — are established products switched on. Brands appearing in a single state and building — Hyppe, Hyde, HQD, Air Bar, and the entire 2022 cohort (Esco Bars, Fume, Avata: 1 state each) — phase in, whether because coverage of that brand improved gradually or because it was a real regional launch.

4. Brand identity vs. documented launch dates

The RMS `product_descr` for essentially all 27 is the generic "SMOKING ALTERNATIVE PRODUCT" / "ANTISMOKING PRODUCT" (Nielsen's post-PMTA disposable label), but the brand-name field and UPC company prefixes identify the products, and most correspond to brands with well-documented pre-2021 market presence:

2021 RMS entrant	product	documented market presence
BIDI	Bidi Stick (Kaival Brands, NASDAQ:KAVL)	launched 2019
KOMGE / MR VAPOR / part of HYPPE	Puff Bar / Puff Plus / Puff Flow (same UPC prefix 08526625)	2019; FDA warning letter July 2020
HYDE	Hyde Edge / Curve / Rave (Magellan Technology)	2020
SHENZHEN HQD TECHNOLOGY	HQD Cuvie	~2018–2019
IGNITE	Ignite V-series (Ignite International, CSE:BILZ)	~2019–2020
SHENZHEN GOLDREAMS	Air Bar (Max / Lux / Box)	~2020
MNGO	MNGO Stick	~2020
STIG	Stig (VGOD)	2018
DINNER LADY	Dinner Lady (UK e-liquid house)	2016 (e-liquid); disposables later
PACHA	Pacha Mama / Pacha SYN (Charlie's Chalk Dust)	2015–2016 (e-liquid); SYN disposables later

2021 RMS entrant	product	documented market presence
VON ERL	Von Erl (device maker; shares UPC prefix 08879690 with BLU)	~2015
AIRIS	Airis / Airis Neo (Airstech)	hardware since ~2016
BO VAPING	Bo (Fontem-adjacent)	~2016
CUE	Cue Vapor System	~2018
UNO	Uno Mas / Uno Bar	~2020
GLAMEE	Glamee / Glamee Nova	~2020
POP	Pop Hit / Pop Hit Pro / "Rare"	~2019

That is ~18 of the 27 (the table covers 20 codes, several as one product) with a product that demonstrably existed and sold elsewhere before its RMS first-appearance date — a coverage case by definition. The remainder (EPIC, ZEO, BUD VAPE, STICK, DUO XTRA, SWITCH MODS, HELO) cannot be tied by name to a documented pre-2021 launch, but all show the same abrupt-onset trajectory (§2) and, where checkable, a broad first-month footprint (§3), so on the internal evidence they read as coverage too, just un-verifiable. As in the brand-dynamics note, these launch dates are from secondary sources and should be checked against company filings / FDA records before citation.

5. Rebrand / reclassification: one product, several brand-codes

Check 4 turns up substantial taxonomy churn — the same physical products are split across multiple RMS brand-codes, which inflates the raw entrant count:

- **UPC prefix 08526625 (the Puff Bar company) is shared by four brand-codes:** PUFF (present pre-2021 but at ~zero tracked mL), KOMGE , MR VAPOR , and part of HYPPE — all with product_descr "PUFF", "PUFF PLUS", or "PUFF FLOW". Puff Bar, the single largest disposable of 2020, thus "enters" RMS in 2021 smeared across three or four codes rather than one.
- **PUUR VAPOR** (prefix 08500197) carries "SAVAGE" UPCs; **SAVAGE** then appears as its own brand-code in 2022 — a split, not a new entrant.
- **HYDE** (prefix 08500254) carries some "HYPPE BAR" UPCs.
- **SHENZEN HQD TECHNOLOGY** and **SHENZHEN GOLDREAMS TECHNOLOGY** use the *manufacturer* name as the brand-code (for HQD Cuvie and Air Bar respectively), where a cleaner feed would have used the product name.
- **VON ERL** shares a UPC prefix with **BLU** — a blu-family device SKU surfacing under the device maker's name.

So of the 27, at least 4–6 are code fragmentation of a product that (a) already existed and (b) partly overlaps another code — they should be merged, not counted as arrivals at all.

6. Entry timing vs. the regulatory dates

First-appearance counts, 2019–2023:

period	new brand-codes
2019 (full year)	3
2020 (full year)	1
Feb–Dec 2020 (post flavored-cartridge enforcement)	0
Sep–Dec 2020 (post PMTA deadline)	0
January 2021	15
Feb–Dec 2021	12
2022 (full year)	10
2023 (full year)	5

If the disposable entry wave were a behavioural response to the February 2020 enforcement (which exempted disposables) or the September 2020 PMTA deadline, new brands would have appeared *through 2020*. Instead 2020 is the single quietest year on the entry margin, and the burst lands entirely in January 2021 — the month the RMS e-cig series switches Nielsen data deliveries. The regulatory shocks are the economic cause of the real-world disposable boom, but the *timing* of the RMS burst is a data-delivery boundary, not a policy date. The Feb-2021-onward trickle (~1/month through 2022–2023) is consistent with a normal low-rate ongoing arrival process (some genuine entry, some continued incremental coverage gains).

7. What this means for the fringe entry process

- **Do not feed the raw first-appearance series into the fringe arrival process.**
Essentially the entire 2021 wave is a coverage/taxonomy discontinuity; treating it as 27 entries (15 of them simultaneous) would overstate the fringe's entry rate several-fold and tie it to a spurious date (a data-delivery boundary).
- **Merge the code fragments first** (Puff family across PUFF/KOMGE/MR VAPOR/part of HYPPE; Puur Vapor/Savage; the manufacturer-named codes). This alone removes 4–6 of the 27.
- **Treat the rest of the 2021 cohort as a one-time "panel onboarding" at January 2021**, not as arrivals. Either (a) drop them from the estimated arrival process entirely, or (b) better, back-date them into the fringe's *incumbent* stock from ~2020, scaling the missing disposable volume to the CDC/IRI all-outlet benchmark (the demand-side notes already use this benchmark). Option (b) also fixes the corresponding under-count of fringe *share* in 2019–2020.
- **The genuine arrival process** — estimated off 2013–2020 plus the 2022–2023 cohort (which shows real single-state launches that ramp) — is a **low-rate, roughly-Poisson process on the order of 2–4 identified fringe brands per year**, with no regulatory-date jump once the coverage artifact is removed. A burst-tolerant or regime-switching arrival process is **not** warranted for the fringe: the apparent burst is in the measurement, not the economics.
- **The regulatory shocks act on the fringe through volume/share, not entry timing.** The Feb-2020 enforcement's visible effect in RMS is the store-panel contraction and the migration of volume toward disposables — a shift in the fringe's *size*, which the model should capture through the fringe's aggregate state, not through a dated entry burst.
- **If a date must be attached to disposable-fringe growth**, use a diffuse "post-2020 regime" rather than Feb-2020 or Sep-2020 specifically — the RMS data cannot date it more precisely than "the disposable fringe was already material by 2020 and the panel caught up in 2021."

Appendix. Method

- Code: `code/manual_refinement/7_entry_wave_store_pass.R` (raw store×week×UPC pass → `input/entry_wave_store_cache.rds`) then `code/analysis/11_entry_wave_coverage_check.R` (all figures and numbers here).
- Panels: `input/bt_month_t2_niccorr_from_full.RData` (brand×type×month), `input/upc_month_niccorr_from_full.RData` (brand×UPC×month), and the raw `raw/cleaned_RMS/full_*` store×week×UPC files for §3.
- Trajectory metric (§2): tracked mL in RMS-months 1–3 ÷ the brand's own median tracked mL over RMS-months 10–24 (or its max if it never reaches month 13).
- Identity/rebrand (§4–5): brand-name text, `product_descr`, and shared 8-digit UPC (GS1 company) prefixes between 2021 entrants and pre-2021 brand-codes.
- Extract boundary (§1, §6): `code/build/1_clean_RMS_from_raw_to_most_detailed.R` (`until2020dir` vs `after2020dir`).
- Launch dates in §4 are secondary-source and should be verified before citation.